Lecture 7 (9/28)

Linear regression model cont. and review

Summary of exam material

Unit 1

- Basics of probability theory: sample spaces, events, joint/marginal/conditional probabilities, independence
- Continuous random variables: Probability density and CDF.
- Distributions you need to know: Bernoulli, Uniform, Normal
- Translating between $\sim$ notation, joint distributions, and code

Unit 2

- Expectation, variance, and the coefficient of variation. Conditional expectation.
- The Normal distribution
- **The linear regression model and its assumptions**
- **Covariance and the relationship to regression coefficient**

Review: Linear regression model

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- Know the assumptions:
  1) Conditional normality,
  2) linearity of conditional expectation (in predictor)
  3) and homoscedasticity of noise.

Review: Regression coefficient and covariance

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- We derived the relation

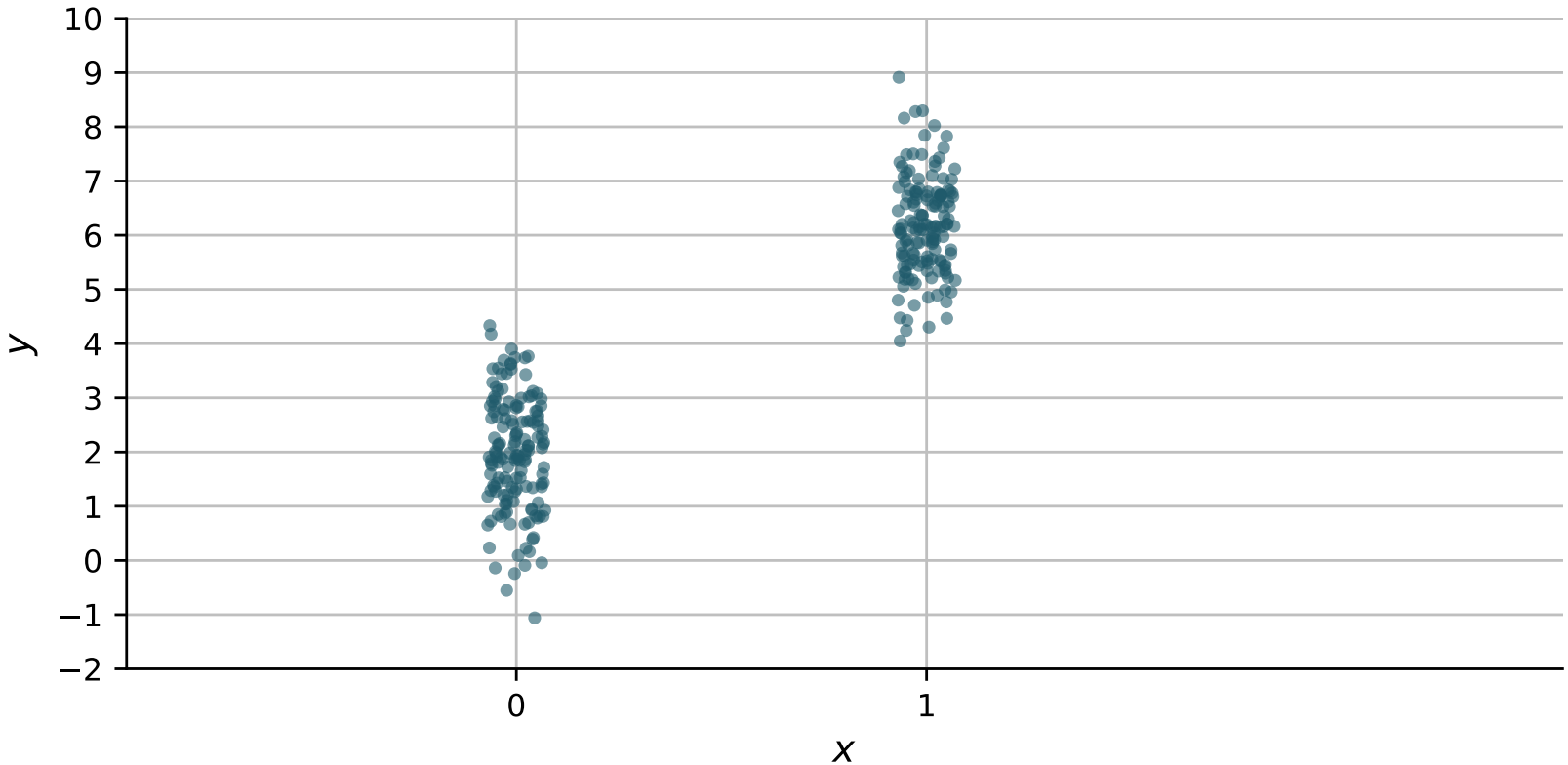
$$\beta_1 = \frac{\mathsf{cov}(X, Y)}{\sigma_X^2}, \quad \sigma_X^2 = \mathsf{var}(X)$$

where

$$\mathsf{cov}(X, Y) = E[XY] - E[X]E[Y]$$

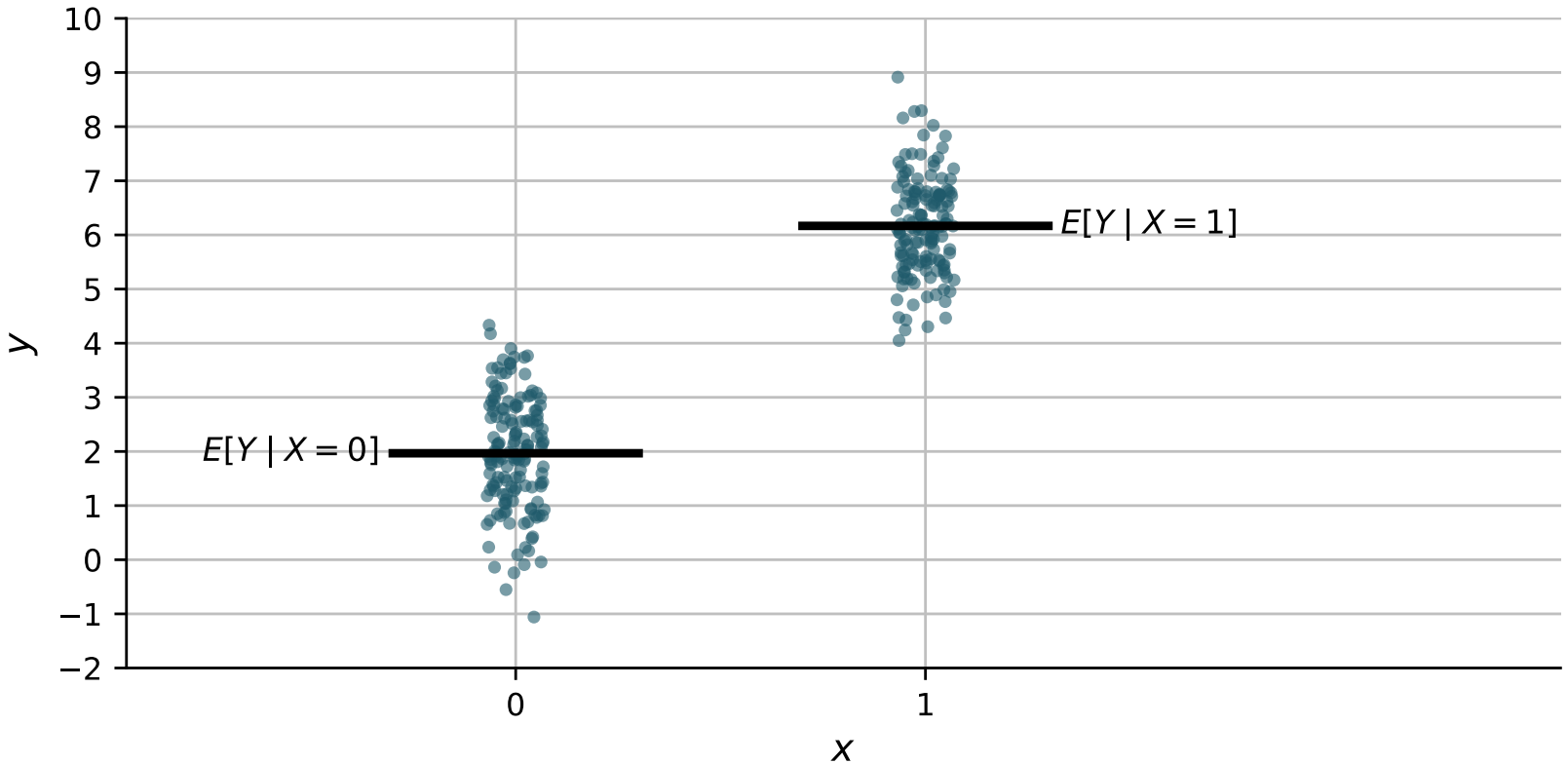
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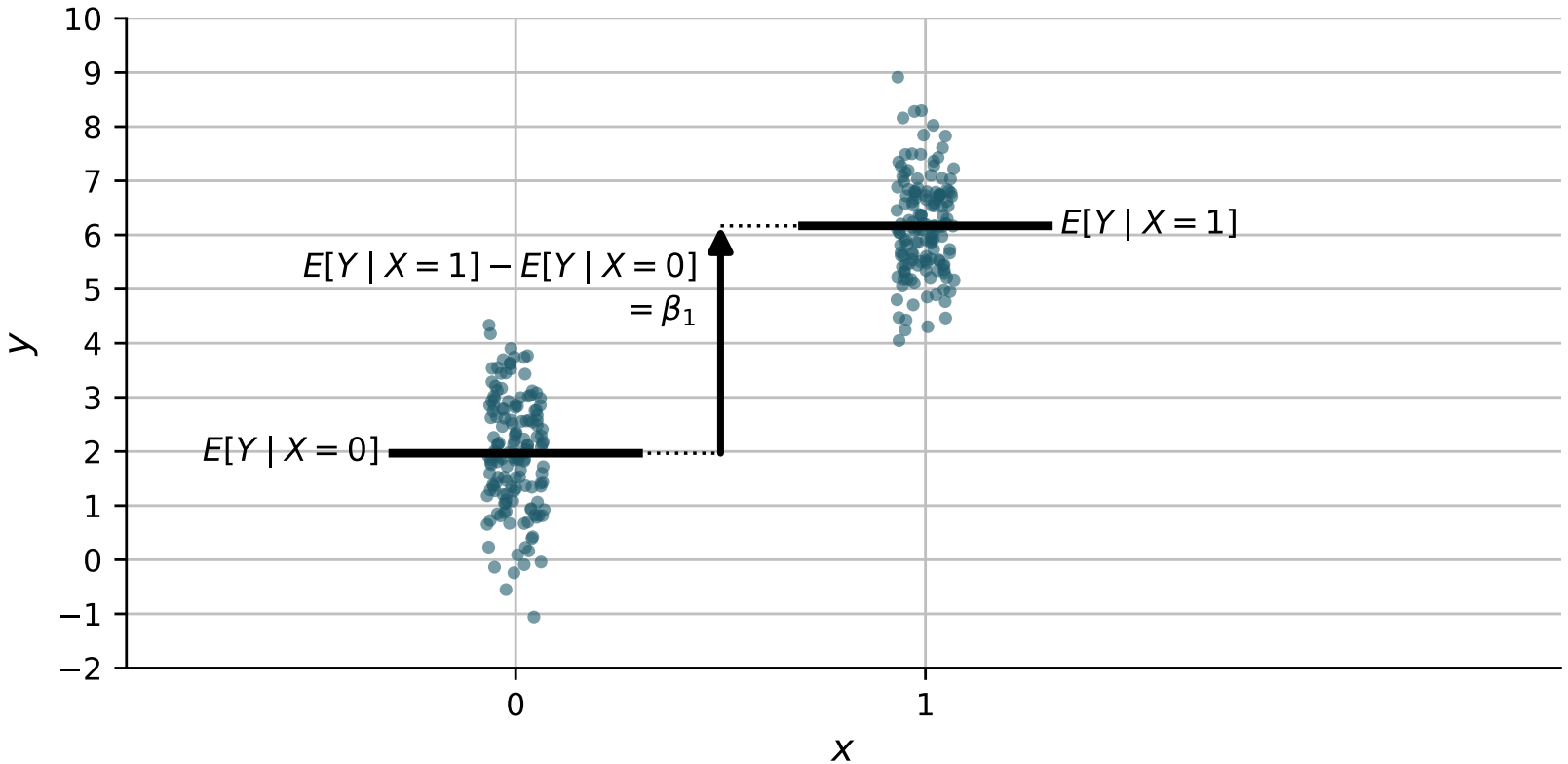
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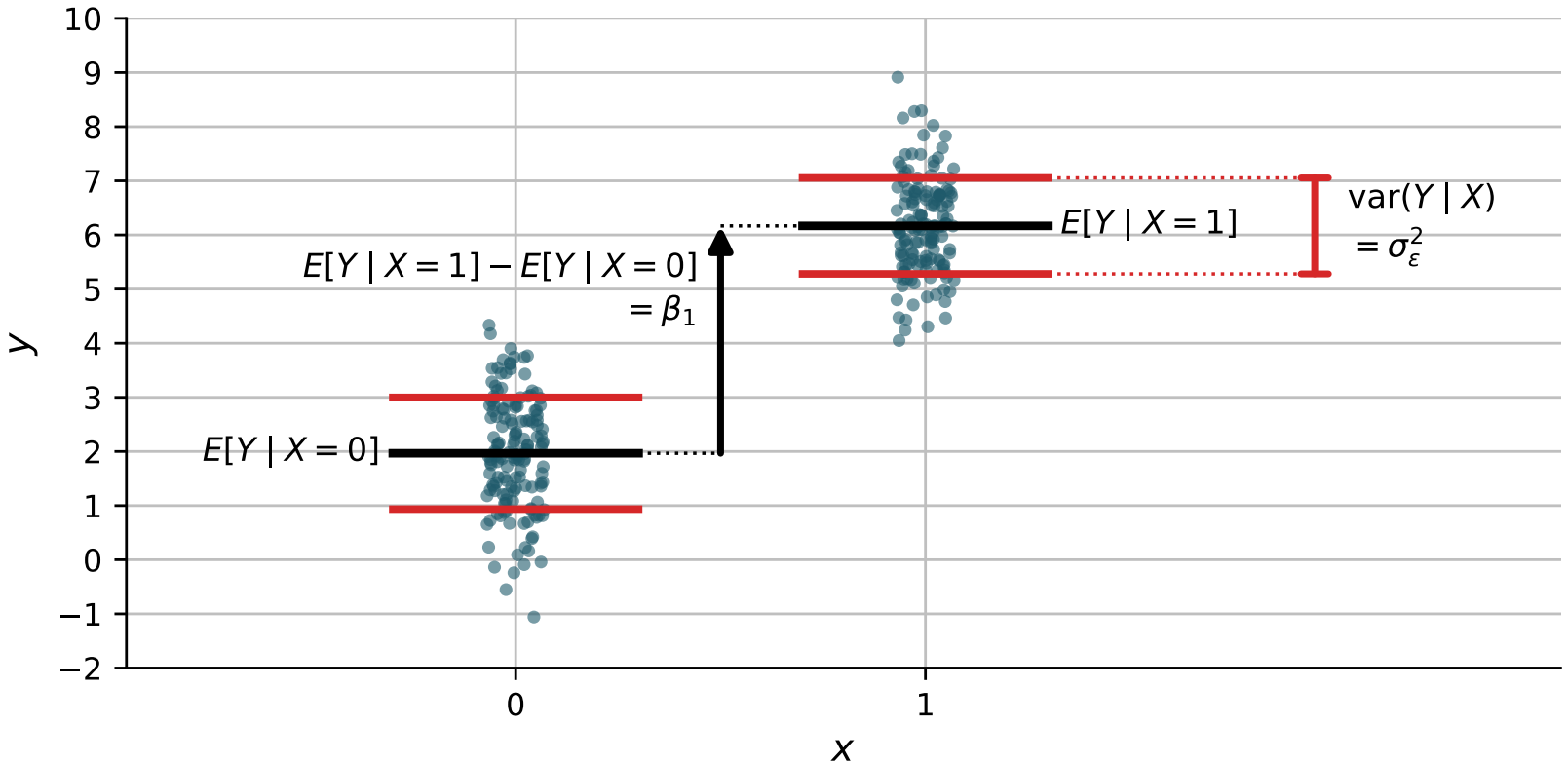
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This decomposition is a special case of the law of total variance (more on this when we discuss ANOVA)

Question

Consider the following code.

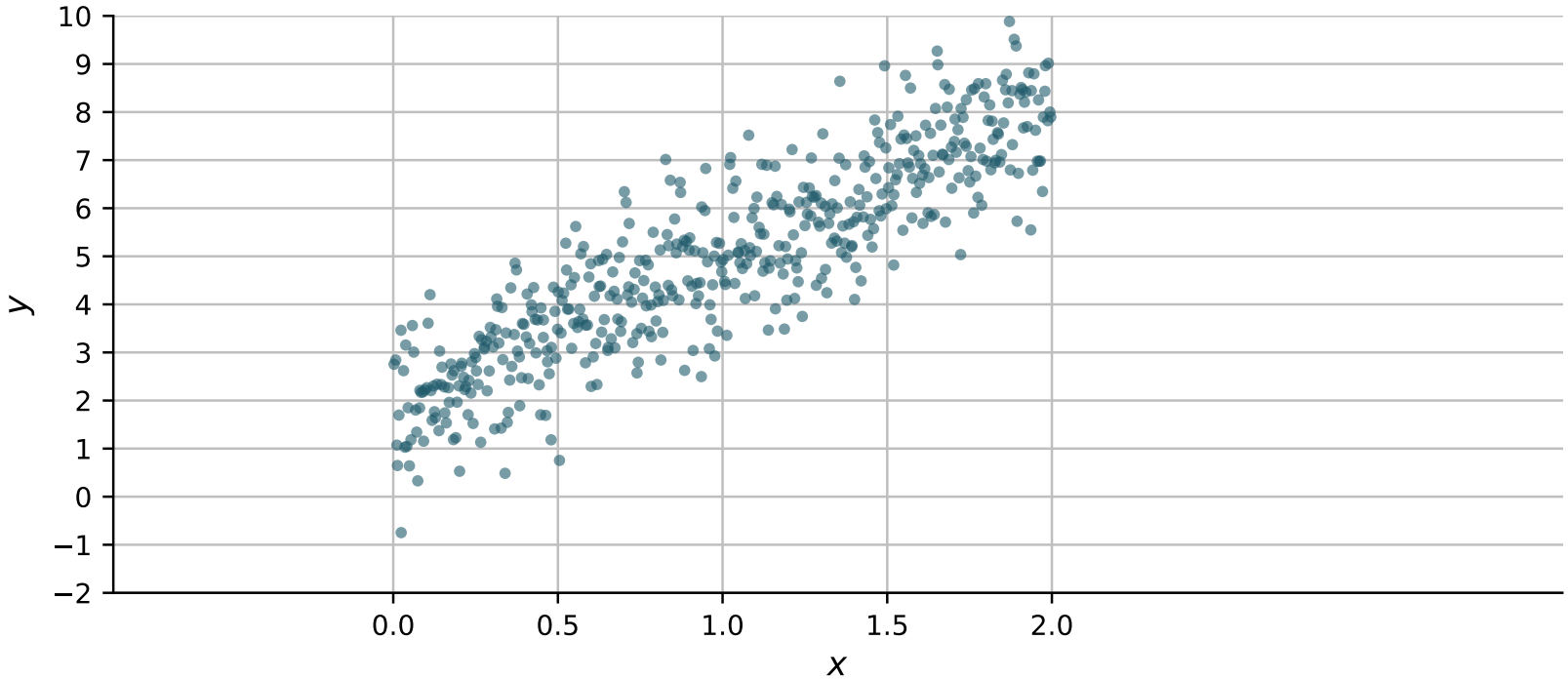
```
n = 100000
x = np.random.uniform(0, 2, n)
y = np.zeros(n)
for i in range(n):
    y[i] = np.random.normal(2 + 3*x[i], 1)

print(np.var(y))
```

1) Write down the probability model being simulated, in $\sim$ notation.
2) What number will `print(np.var(y))` show, approximately?

Question

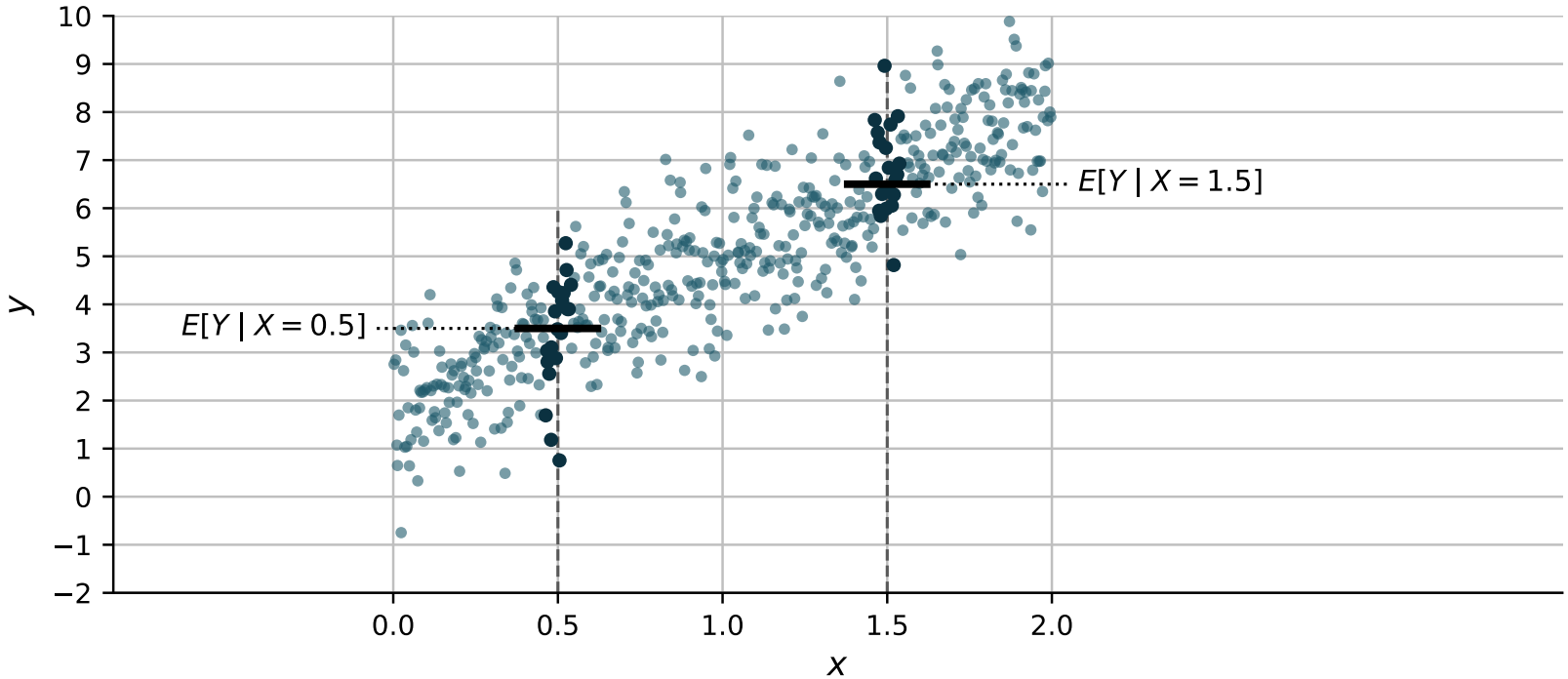
The plot below shows samples (x_i, y_i) from the same model as the previous slide.



Estimate $\mathsf{cov}(X, Y)$, σ_ϵ^2 and σ_Y^2 from the plot.

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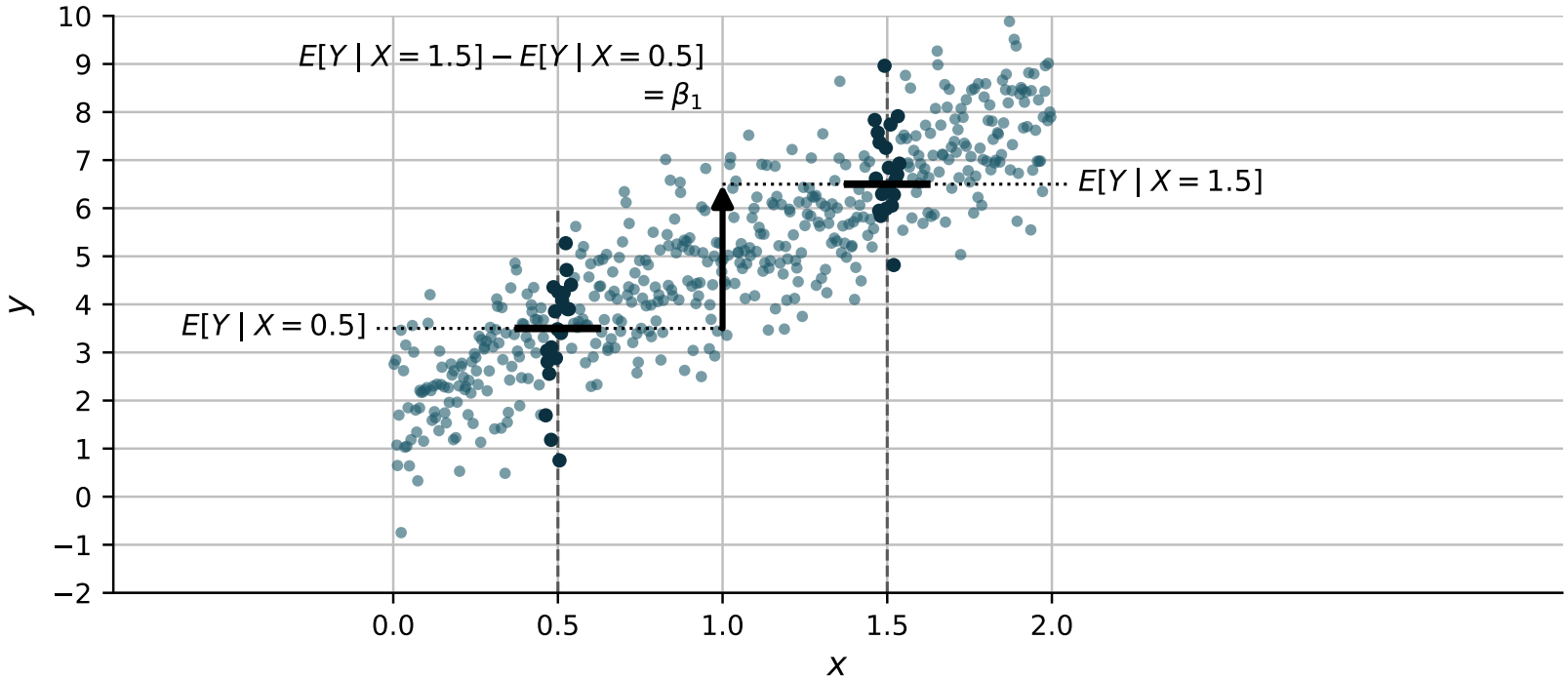
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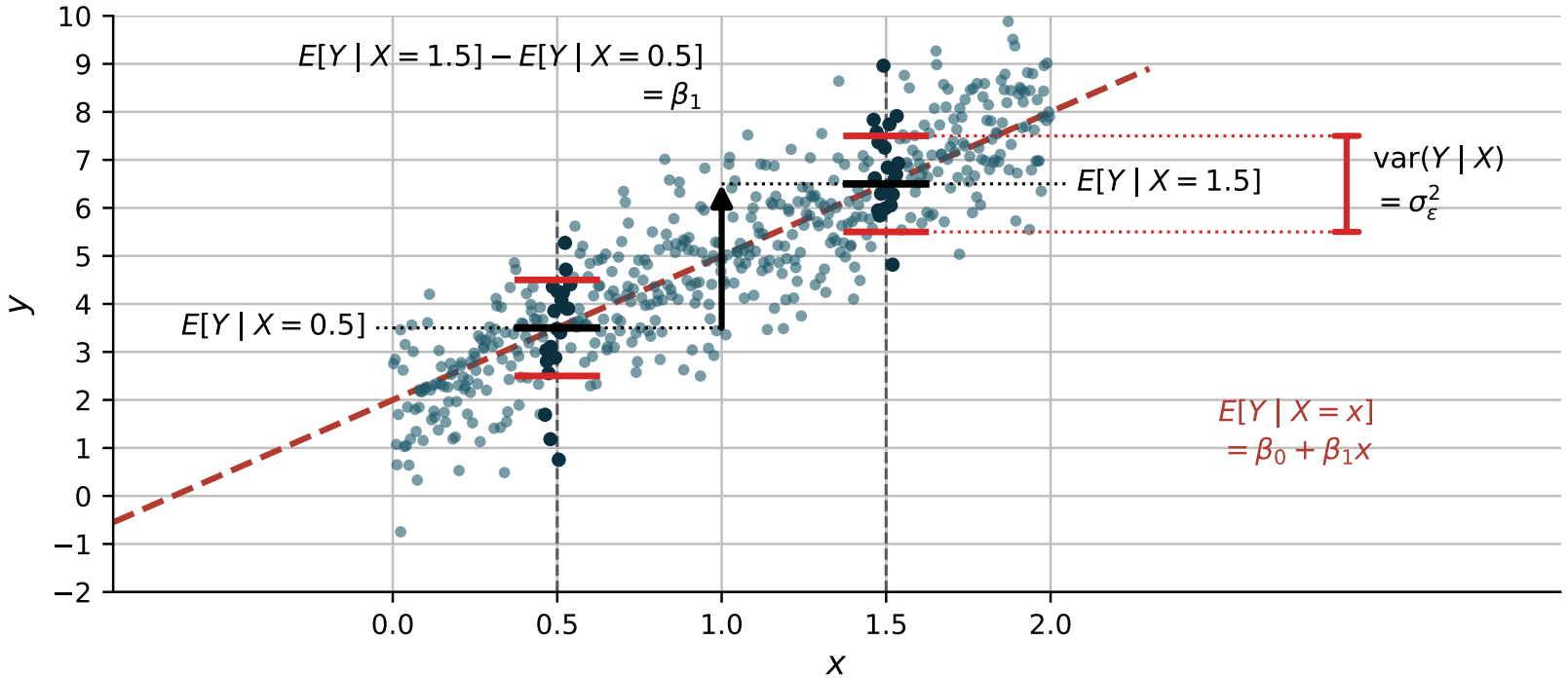
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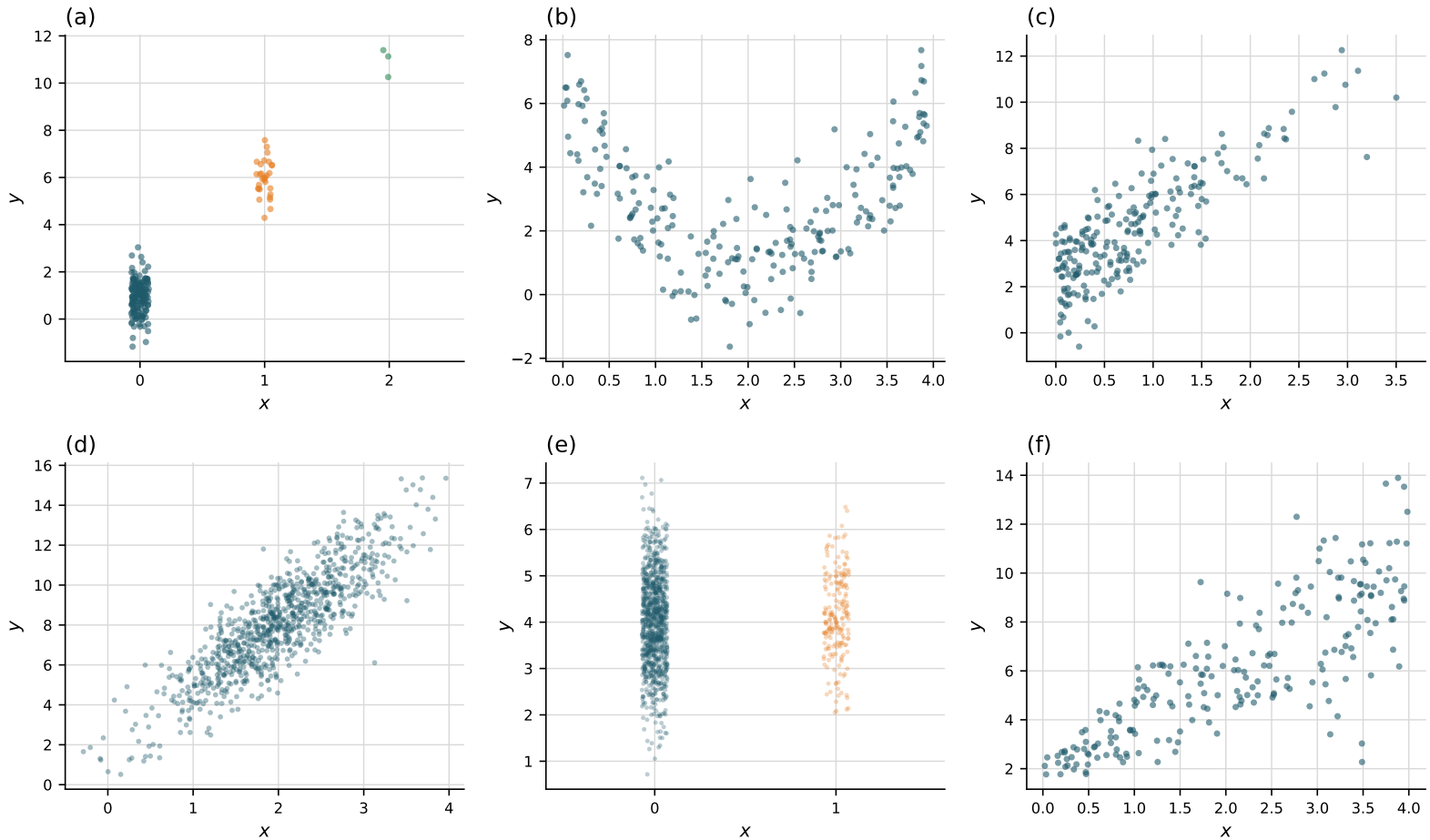
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Question

Which of the following scatter plots could have been generated by a linear regression model?



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$$Y \mid X \sim \text{Bernoulli}(X/3 + 1/3)$$

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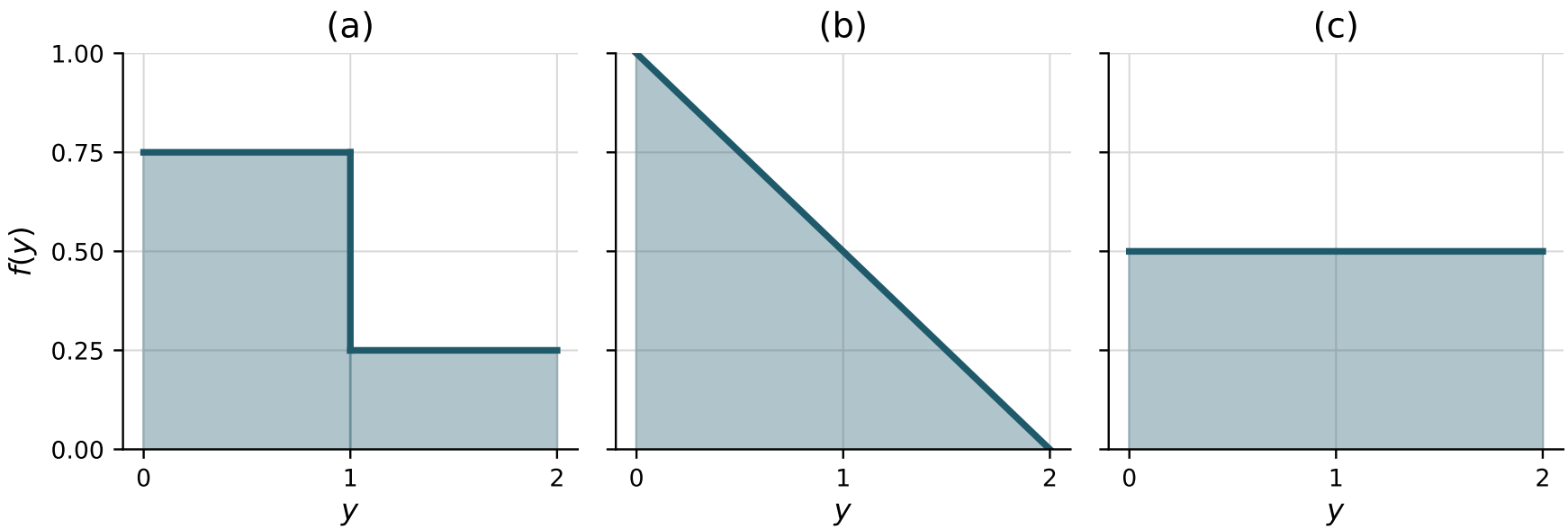
$$E[Y \mid X = 1] - E[Y \mid X = 0] = \frac{\text{cov}(X, Y)}{\text{var}(X)}?$$

- Is this always true when $E[Y|X]$ is linear in X?

Question

A lottery scratch ticket has a base cash prize (Y – measured in hundreds) which is equally likely to be any value between 0 and \$100 but the ticket has a 50/50 chance of revealing a doubler symbol (X), which doubles the value.

- Write this down as a conditional model for X and $Y \mid X$.
- Which of the following could be the marginal density of Y?



Question

```
n = 100000
x = np.random.choice([1,2,3], size=n)
m1, m2, m3 = x==1, x==2, x==3
y = np.zeros(n)
y[m1] = np.random.choice([1,2,3], np.sum(m1), p=[.6,.3,.1])
y[m2] = np.random.choice([1,2,3], np.sum(m2), p=[.2,.6,.2])
y[m3] = np.random.choice([1,2,3], np.sum(m3), p=[.1,.3,.6])
result = np.mean(x[y==1])
```

What is the value of `result`?

Question

```
n = 100000
p = [.1, .25, .3, .35]
out = np.random.choice([1,2,3,4], p=p, size=n)

y = np.zeros(n)
x = np.zeros(n)
x[out==2] = 1
x[out==4] = 1
y[out==2] = 1
y[out==3] = 1
```

Write down a conditional model for a pair of Bernoulli random variables X and Y which is sampled by this code. That is, find the distribution of X and $Y \mid X$.

Question

We are given a dataset in the form of two numpy arrays, `x` and `y`. Each index represents the outcome of an election for Governor and Senate in a different state, with Democrat, Independent and Republican coded as 0, 1 and 2 respectively. A political scientist has the hypothesis that the two are independent. Describe a procedure for testing whether this is true in Python[a]?

[a] without using out of the box hypothesis tests we haven't learned about

Question

Consider the following:

- 10% of all applicants to a particular graduate program are admitted.
- Among admitted students, 90% had a GPA above 3.8
- Among applicants who were *not* admitted, 30% had a GPA above 3.8

What is the probability an applicant with a GPA above 3.8 is admitted? Write Python code to sample from the joint distribution of A, G (Bernoulli variables representing whether a student is admitted and has GPA above 3.8 respectively).

Solutions

1) Binary-predictor plot: group means ≈ 2 and 6, so $\beta_1 \approx 4$; $\sigma_X^2 = \frac{1}{4}$ and $\sigma_\epsilon \approx 1$, so $\sigma_\epsilon^2 \approx 1$. Hence $\mathsf{cov}(X, Y) \approx \beta_1 \sigma_X^2 = 1$ and $\sigma_Y^2 \approx \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = 5$.

2) $X \sim \mathsf{Uniform}(0, 2)$, $Y \mid X \sim \mathsf{Normal}(2 + 3X, 1)$. Since $\sigma_X^2 = \frac{(2-0)^2}{12} = \frac{1}{3}$: $\mathsf{var}(Y) = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = 9 \cdot \frac{1}{3} + 1 = 4$, so `np.var(y)` prints ≈ 4.

3) Uniform-predictor plot: reading off $E[Y \mid X{=}0.5] = 3.5$ and $E[Y \mid X{=}1.5] = 6.5$ gives $\beta_1 \approx (6.5 - 3.5)/1 = 3$; $\sigma_\epsilon \approx 1$. Hence $\mathsf{cov}(X, Y) \approx \beta_1 \sigma_X^2 = 3 \cdot \frac{1}{3} = 1$, $\sigma_\epsilon^2 \approx 1$, and $\sigma_Y^2 \approx \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = 4$ – matching the previous problem, since it's the same underlying model.

4) Six-panel check: (a) yes (b) no (c) yes (d) yes (e) yes (f) no

5) $X \sim \mathsf{Bernoulli}(\frac{1}{2})$, $Y \mid X \sim \mathsf{Bernoulli}(X/3 + 1/3)$: $\mathsf{cov}(X, Y) = \frac{1}{12}$. Yes: $E[Y \mid X{=}1] - E[Y \mid X{=}0] = \frac{1}{3} = \mathsf{cov}(X, Y)/\mathsf{var}(X)$. Whenever $E[Y \mid X] = \beta_0 + \beta_1 X$ the identity holds.

Solutions (cont.)

6) Lottery ticket: $X \sim \text{Bernoulli}(\frac{1}{2})$, $Y \mid X{=}0 \sim \text{Uniform}(0,1)$, $Y \mid X{=}1 \sim \text{Uniform}(0,2)$ (doubling the range, in hundreds of dollars); the marginal density is panel (a).

7) Reverse-engineered $\{1,2,3\}^2$ model: `result` estimates $E[X \mid Y{=}1] = \frac{13}{9} \approx 1.44$.

8) Reverse-engineered Bernoulli pair: $X \sim \text{Bernoulli}(0.6)$; $Y \mid X{=}0 \sim \text{Bernoulli}(0.75)$; $Y \mid X{=}1 \sim \text{Bernoulli}(\frac{5}{12})$.

9) Election independence: compute $\hat{P}(X{=}i, Y{=}j)$ for each of the 9 pairs and compare to $\hat{P}(X{=}i)\hat{P}(Y{=}j)$; to judge whether any gap is bigger than noise, repeatedly shuffle the `y` array relative to `x` (breaking any true association) and recompute the same discrepancy each time, then check whether the discrepancy in the real data is unusually large compared to this shuffled/null distribution.

10) GPA/admissions: $P(\text{Admit} \mid \text{GPA} > 3.8) = \dfrac{0.9 \times 0.1}{0.9 \times 0.1 + 0.3 \times 0.9} = \dfrac{0.09}{0.36} = 25\%$.